S26 Internship Presentation

Cristopher Miranda | Payload & Mechanisms Intern

Introduction

About Me

Personal Info dump

- Born and raised in Norcross, GA (Metro ATL)
  - the payload team is disproportionately Georgian..?
- Rising senior studying Mechanical Engineering
- Before Varda
  - Thermal engineering intern, Starfish Space
  - Undergraduate researcher, MIT Space Propulsion Lab
  - Vice-President, SEDS@MIT (space fan club)
  - Solid Propulsion RE, MIT Rocket Team
- Interests
  - Hiking (thanks Seattle)
  - Eating food ($1.00 street tacos)
  - Listening to the same song on repeat
  - Machining, especially on the lathe but the mill grew on me
  - Huge Kurt Vonnegut fan – I met one of the directors of the Kurt Vonnegut museum this spring!
  - Taking the Metro

Photos

Awesome spring!

Previous Projects

MIT Clubs & Classes + Previous Professional Experience

- N2O + Paraffin Hybrid Rocket Design - Colibri UP (Mexico City, MX)
  - Designed an IREC-capable rocket with an apogee of 10,000 ft to support a university rocketry team in Mexico. I also mentored their team on designing solid motors.
- Solid Rocket Motor Design – MIT Rocket Team (Cambridge, MA)
  - Design, manufacture, integrate, and test of solid rocket motor for IREC
- TVAC Chamber Design – Starfish Space (Tukwila, WA)
  - Bring up of TVAC chamber with very similar architecture to the chambers here (Chiller > Liquid Cooled Plate > TECs > Optical Board (I promise I didn't steal the idea))
- Mass Manufactured Yoyos – MIT 2.008 (Cambridge, MA)
  - Designed press fit components and IM molds for a gumball themed yo-yo as part of a six-person team
  - We created 50 yoyos in total!
- Thruster Emulator Manufacturing – MIT Space Propulsion Lab (Cambridge, MA)
  - Micro-machining of PEEK thruster emulators for a 3U CubeSat

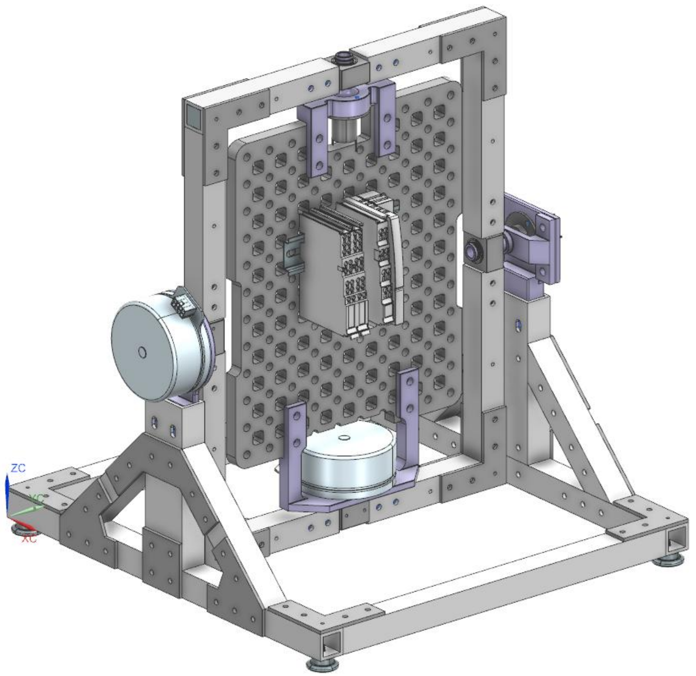


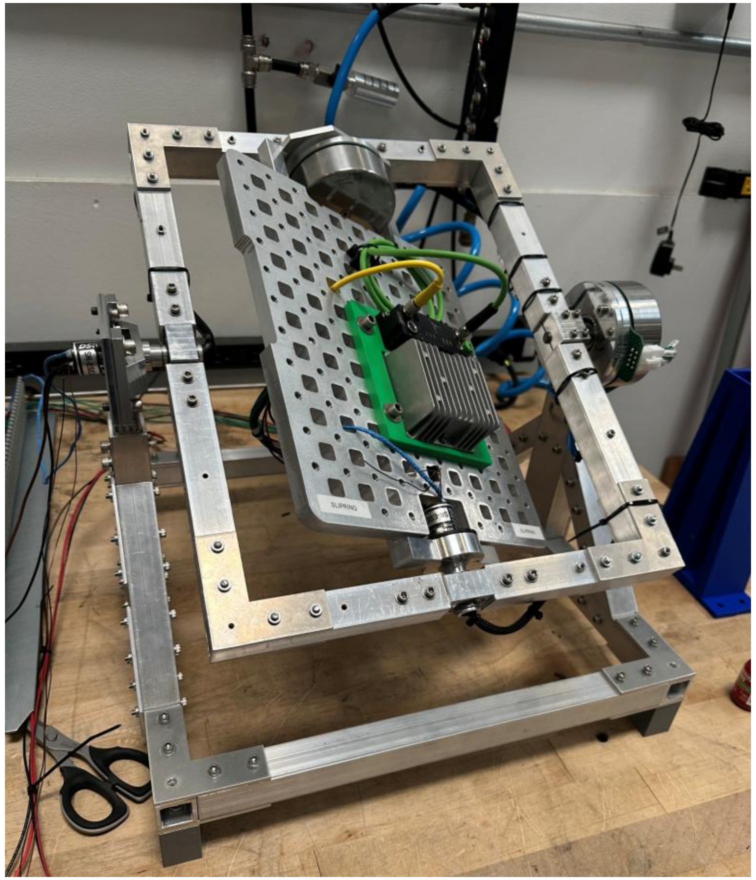
Technical Project Overview - RPM

Design

General Overview

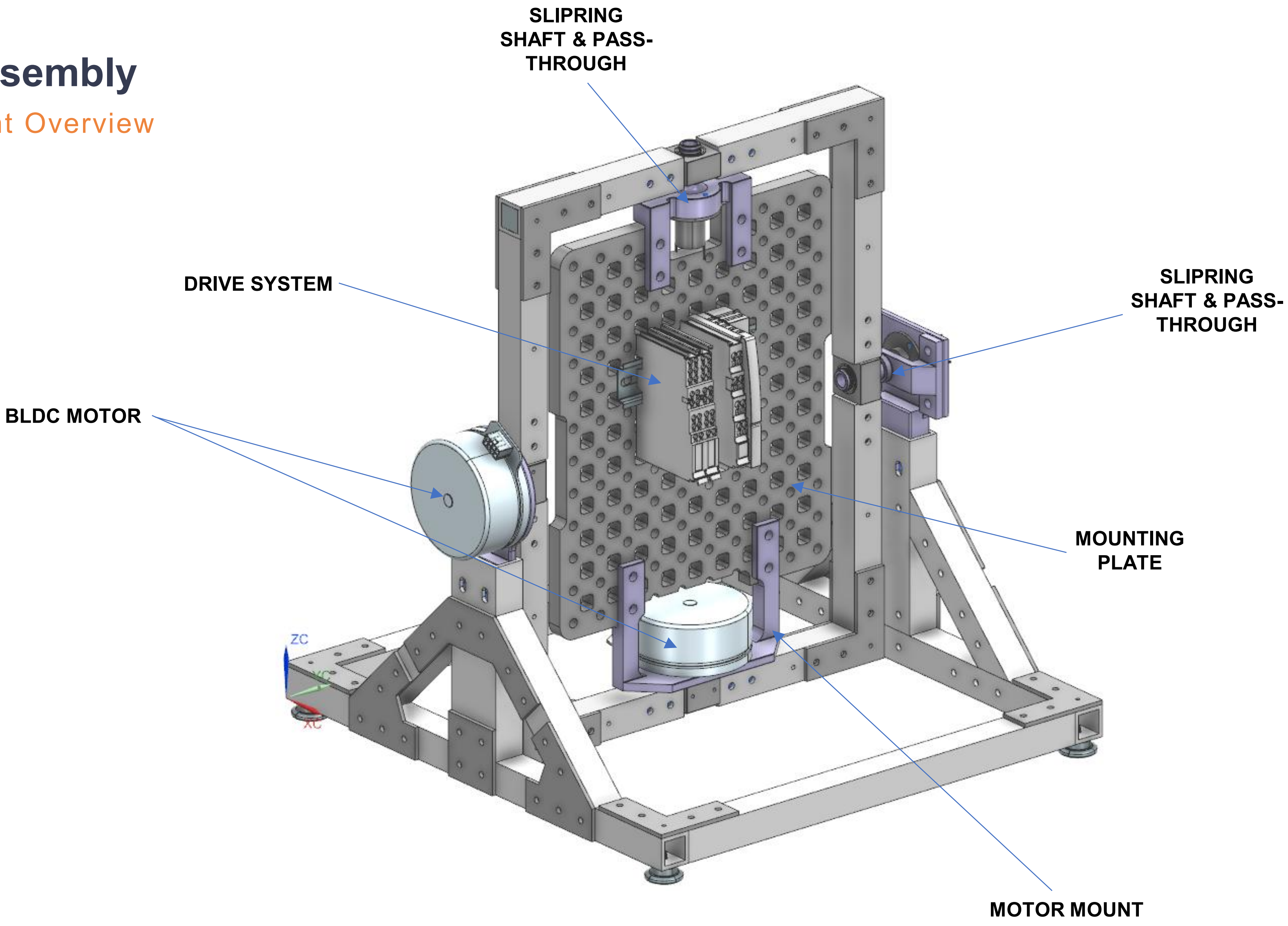
- Two independent, perpendicular rotating frames
- **Total Mass** 11 kg
- **Speed** 0-60 rpm
- 7 unique machined components
- 8 unique laser cut components





RPM Assembly

Component Overview



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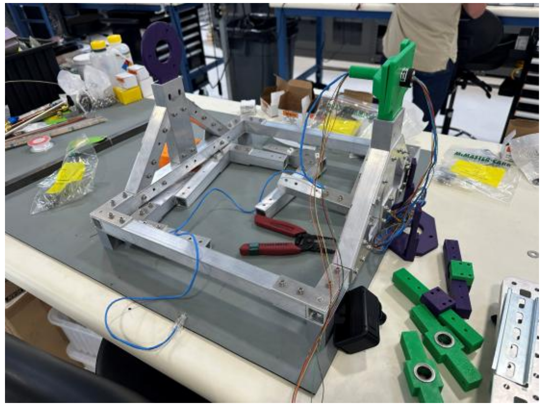
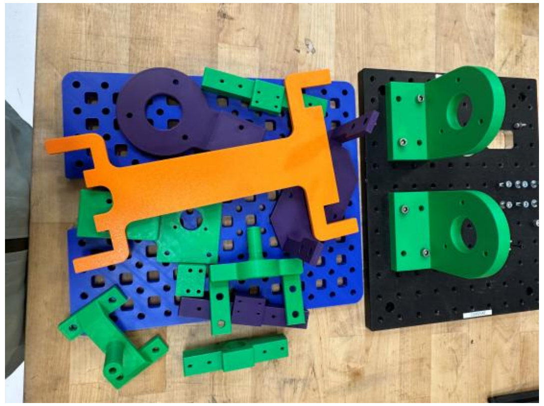
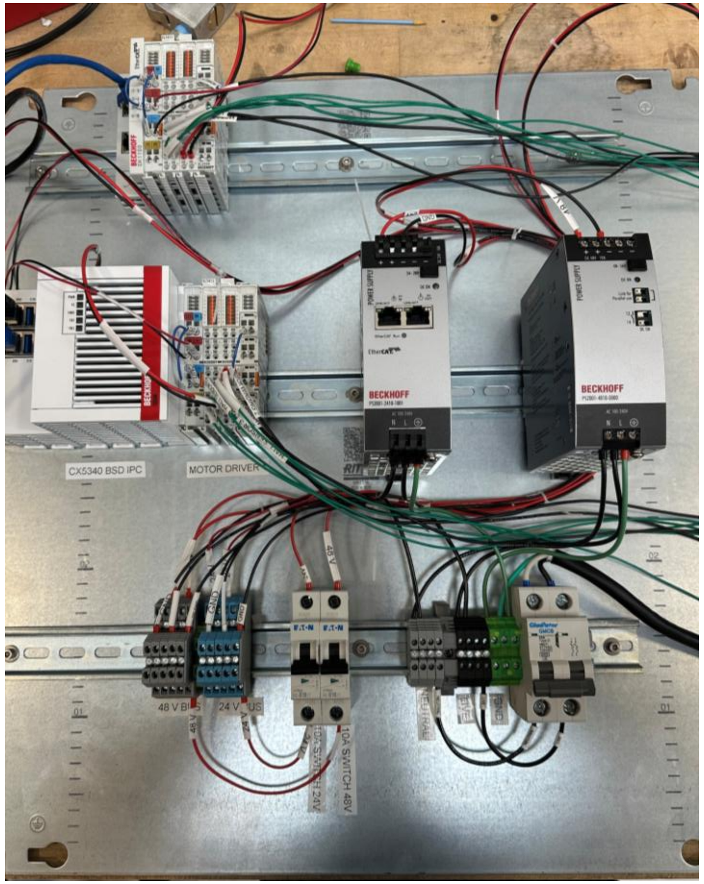
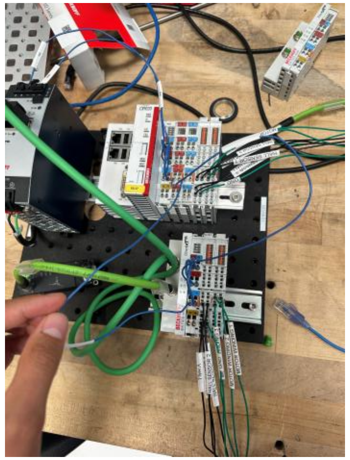
Build Photos

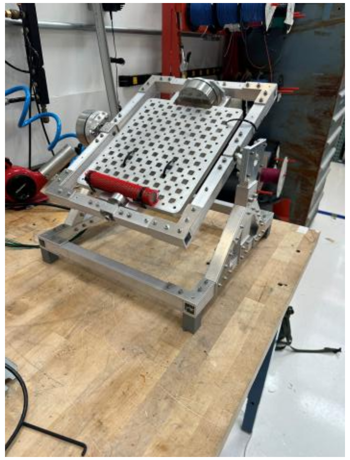
3D Printed Parts, Remachining, Beckhoff, and more…

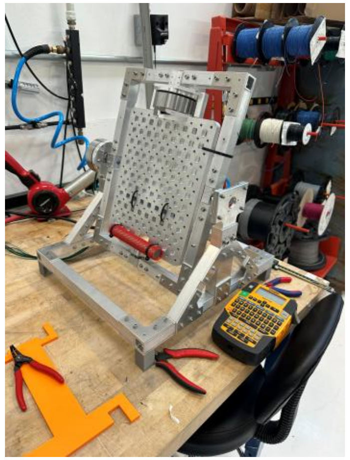


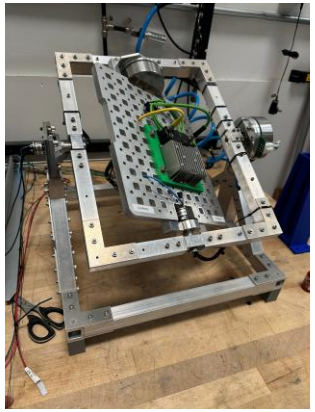












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Motor Selection

Maxon EC90 BLDC (260W)

Selection

- Motor selected due to combination of high torque, favorable packaging, and manageable current requirement
  - $T_{nominal}$ = 964 mNm
  - $I_{nominal}$ = 4100 mA
  - Due to space constraints, a short and wide motor was more suitable
  - Originally, a similar motor at 24V was selected, but I pivoted because the nominal current was too high (exceeded output of driver)

Speed, Position Control

- The selected motor has an option for 4096 CPR encoder
- The configuration I ordered is encoder-less and relies on hall sensors for feedback
- This added challenges to tuning the motor, and achieving precise control at low speeds

Current motor (hall sensor only)

Ideal Motor (hall sensor + 4096 CPR encoder)

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Motor Tuning

Parameter Tuning in TwinCAT

P.I.D Controller

- **P** gain controls the short-term errors (current distance to target)
- **I** gain controls the error as it has accumulated over time
- **D** gain predicts future error by observing the rate it is changing
    - Less useful here since we were tuning very noisy data
- The two motors required very different tuned parameters, which makes sense given their difference in loading conditions

Step 1: Optimization of the velocity controller

1. Ensure that the voltage constant of the motor 8011:31 "Voltage constant" is set correctly.
2. Set the parameter 8010:60 "Sensorless max. acceleration" to a lower value, otherwise the velocity jump can become too big. e.g. 2000 °/s².
3. Reduce the parameter 8010:5E "Sensorless offset voltage scaling" to approx. 50 ... 80%.
4. Set the parameter 8010:17 "Position loop proportional gain" to 0.
    → The position controller is disabled.
    → Interference of the position controller with the velocity controller is prevented.
5. Set the parameter 8010:5B "Velocity loop proportional gain (voltage mode)" to 0.
6. Set the parameter 8010:5C "Velocity loop voltage feed forward gain (voltage mode)" to 100%.
7. Set the parameter 8010:5F "Sensorless observer bandwidth" to a lower value. e.g. 50 Hz.
8. Gradually increase the parameter 8010:5F "Sensorless observer bandwidth" until oscillation occurs. Then reduce by 50%.
9. Configure the integral component of the velocity controller in parameter 8010:5A "Velocity loop integral time (voltage mode)" to be rather sluggish.
10. Gradually increase the proportional component 8010:5B "Velocity loop proportional gain (voltage mode)" until the actual velocity in the scope begins to oscillate.
11. Reduce the proportional component by 20%.
    The 20% serves as a control reserve for abrupt movements.
12. If the velocity overshoots: slightly reduce the parameter 8010:5C "Velocity loop voltage feed forward gain (voltage mode)".
13. If necessary, increase the parameter 8010:60 "Sensorless max. acceleration" back to the required dynamics.

Beckhoff guidance on tuning velocity controllers

	Observation (at default parameters)	Action	Result
Motor 1 (well balanced, higher rotary inertia)	Very high overshoot, periodic oscillation (speeds up very quickly, changes direction)	Negate the I term (increase), increase P gain	Significantly improved stability (±4 rpm)
Motor 2 (less balanced, lower rotary inertia)	Moderate overshoot, no discernable oscillation	Keep low I term, increase P gain	Significantly improved stability (±3 rpm)

Motor Tuning

Parameter Sheets

- Operating in voltage mode (a result of running in velocity mode), there are a few knobs/parameters to turn
  - **<u>8010:5A</u>** Velocity loop integral time (voltage mode)
  - **<u>8010:5B</u>** Velocity loop proportional gain (voltage mode)
  - 8010:5C Velocity loop forward gain (voltage mode)
  - 8010:5E Sensorless offset voltage scaling
  - 8010:5F Sensorless observer bandwidth
- Importantly, there is no automatic PID sensing tool for voltage mode on the motor driver.
- The first two parameters offer the greatest leverage over motor behavior

Overshoot

8010:59	Error suppression mask	RW	0x00000000 (0)	
8010:5A	Velocity loop integral time (voltage mo...	RW	0x00000032 (50)	0,1 ms
8010:5B	Velocity loop proportional gain (voltag...	RW	0x00000BB8 (3000)	µV/(°/s)
8010:5C	Velocity loop voltage feed forward gai...	RW	0x00 (0)	%
8010:5E	Sensorless offset voltage scaling	RW	0x0032 (50)	%
8010:5F	Sensorless observer bandwidth	RW	0x00FA (250)	Hz
8010:60	Sensorless max acceleration	RW	0x000007D0 (2000)	°/s^2
8010:61	Cogging torque compensation	RW	00 00 00 00 00 00 00 00 00 00 00 00 ...	
8010:62	Position loop deadband window	RW	0x00000000 (0)	
8010:63	Find commutation time	RW	0x0009 (9)	0,1 s
8010:64	Commutation type	RW	Six-Step with hall (2)	

Motor 1 (External)

8010:59	Error suppression mask	RW	0x00000000 (0)	
8010:5A	Velocity loop integral time (voltage mo...	RW	0x000186A0 (100000)	0,1 ms
8010:5B	Velocity loop proportional gain (voltag...	RW	0x000003E8 (1000)	µV/(°/s)
8010:5C	Velocity loop voltage feed forward gai...	RW	0x0A (10)	%
8010:5E	Sensorless offset voltage scaling	RW	0x0032 (50)	%
8010:5F	Sensorless observer bandwidth	RW	0x00FA (250)	Hz
8010:60	Sensorless max acceleration	RW	0x000007D0 (2000)	°/s^2
8010:61	Cogging torque compensation	RW	00 00 00 00 00 00 00 00 00 00 00 00 ...	
8010:62	Position loop deadband window	RW	0x00000000 (0)	
8010:63	Find commutation time	RW	0x0009 (9)	0,1 s
8010:64	Commutation type	RW	Six-Step with hall (2)	

Motor 2 (internal)

Motor Resolution

Encoder vs Hall-Sensor Feedback

Hall-Sensor Resolution

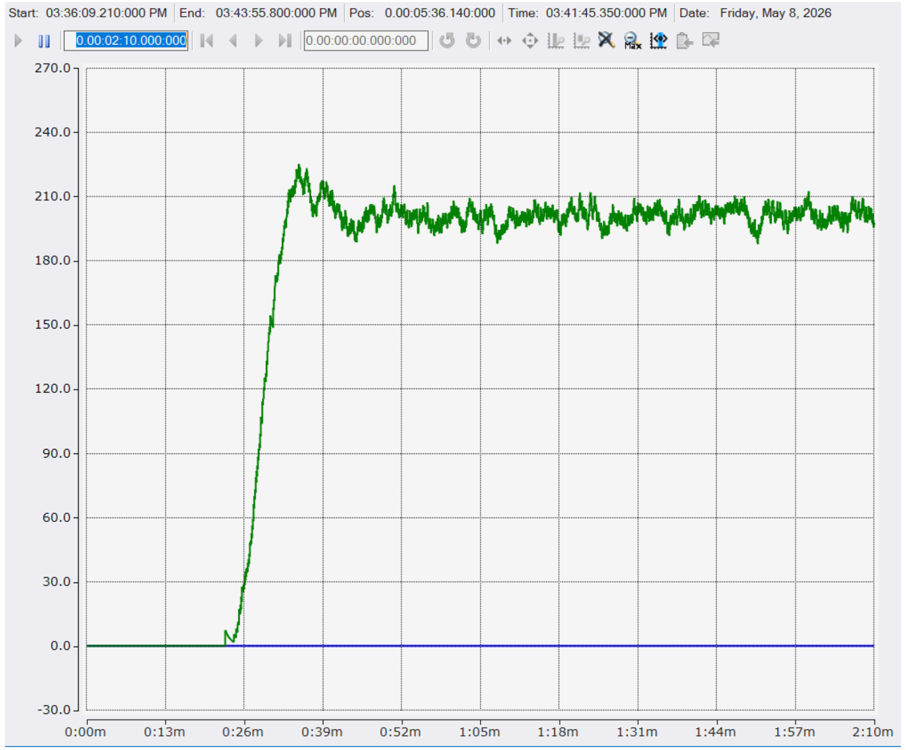
- x3 hall sensors
- 11 pole pairs on motor
- ~5.45 degrees between steps

Encoder Resolution

- 4096 CPR (standard add-on)
- ~.087 degrees between steps (62x improvement)

Implications

- Finely tuned control parameters still experience considerable overshoot, especially when subjected to imbalanced load
- Clear oscillatory behavior
- Severely limits the ability to accurately operate the motor at low speeds
- Lower limit for motor ~25 rpm, well above the lower range of speeds for RPMs



Electrical Work

Beckhoff, Terminal Blocks, and 18 AWG wire…

Power Supplies

- 24V PSU (supply, peripheral power)
- 48V PSU (input power for motors)

Terminal Blocks

- Distribute up to 10A at 48V to BLDC motor drivers power input
- Distribute up to 10A at 24V to supply and peripheral input for PLCs

Electrical Relays/Fuses

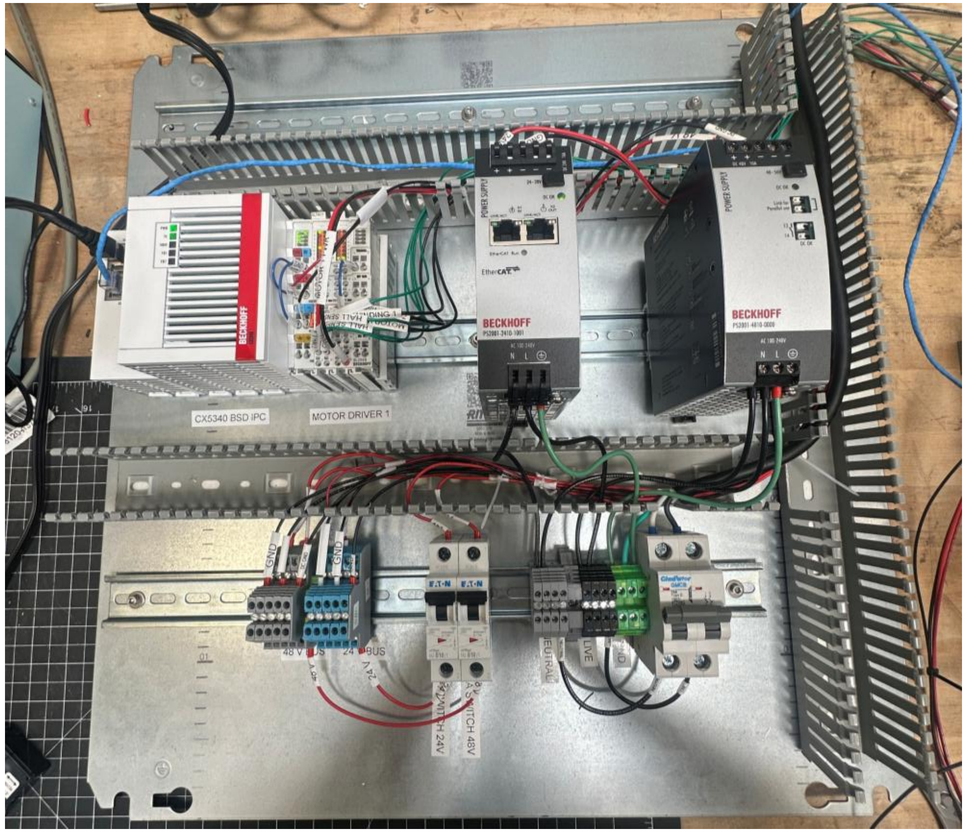
- 10A safety built into both PSUs
- 10A fuse and switch along 24V, 48V lines before terminal blocks

Slip Rings

- Transmit power and data at 48V <10A across rotary joint

Beckhoff Modules

- IPC > Coupler > Driver > I/O



Slip Ring Selection

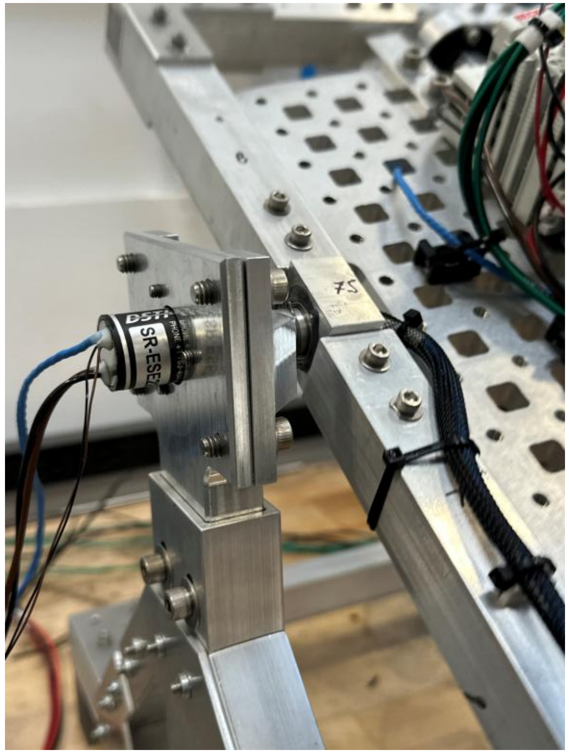
Moog 12A 4 circuit Slipring with Ethernet

Selection

- Small form factor, short lead time, good documentation, and satisfactory current capability
- Circuits Available
    - 1x Ethernet (only 4 wires, not full 8)
    - 1x 2A (26 awg)
    - 1x 10A (18 awg)

Troubleshooting

- The 26 awg wire on the 2A circuit is delicate (easy to bend, hard to crimp, not stiff)
- The large, stiff wires had a larger bend radius than expected
- After some rounds of testing, the 2A circuit on both sliprings failed, likely due to an internal short



Design and Build Challenges

Overcoming Hurdles

- Early miss on motor selection (encoder-less)
- Heavy mounting plate with little done to reduce mass or reduce footprint
  - Required machining some pockets in-house to provide additional clearance for harnesses
- Premature failure on 2A circuit through slipring
  - Still missing root cause. Likely due to overcurrent from PLC bus power draw causing an internal short
  - The issue of running 24V and 48V through a single circuit on the slipring was solved by using a buck converter on the rotating stage
- Bearing housing oversized (.002”), requiring retention compound and staking on bearing flanges
- Underestimated challenges with placing drivers on the inner rotating frame
  - Increased rotating mass considerably
  - Introduced additional failure points (quick disconnects, extra slipring)

R.I.P slip ring

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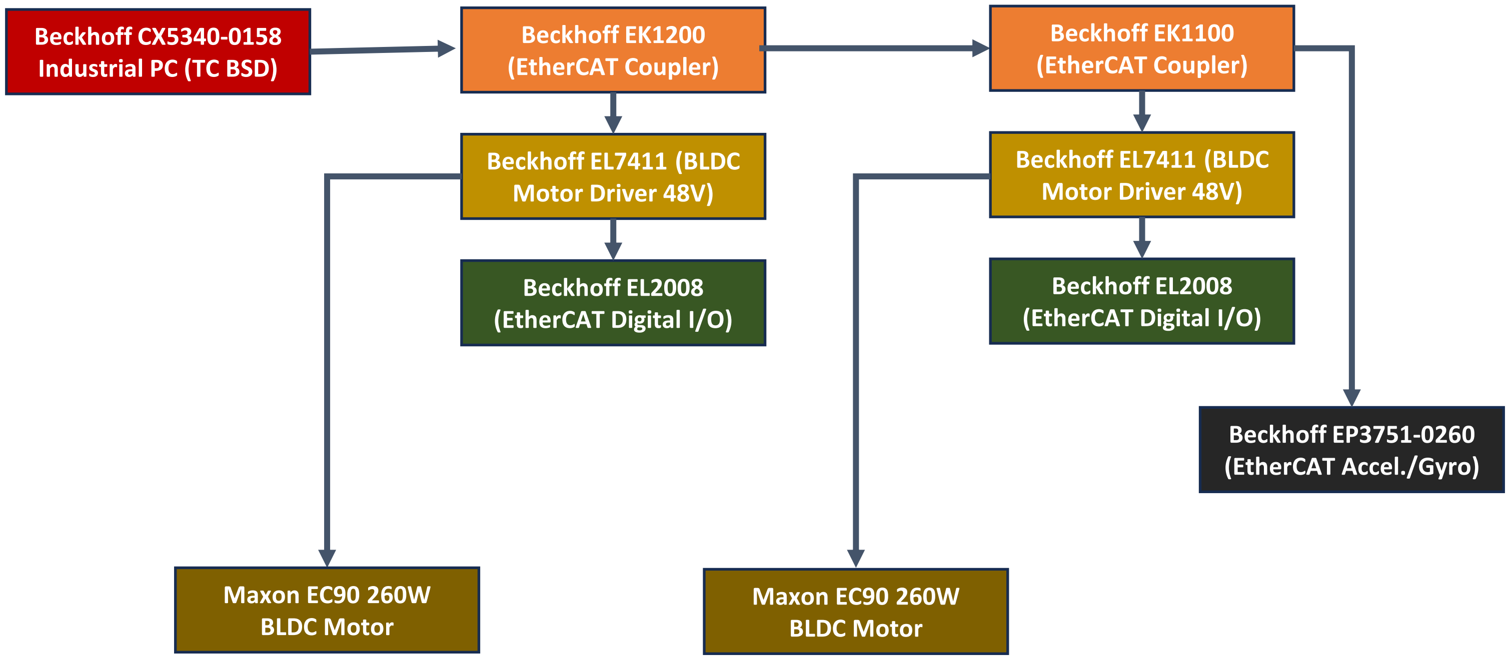
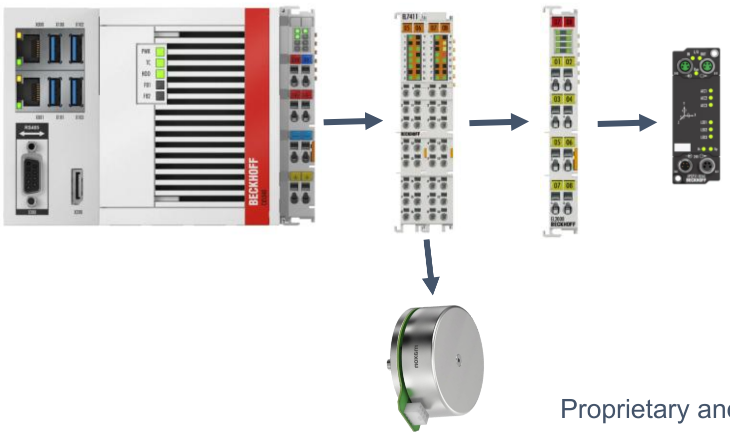
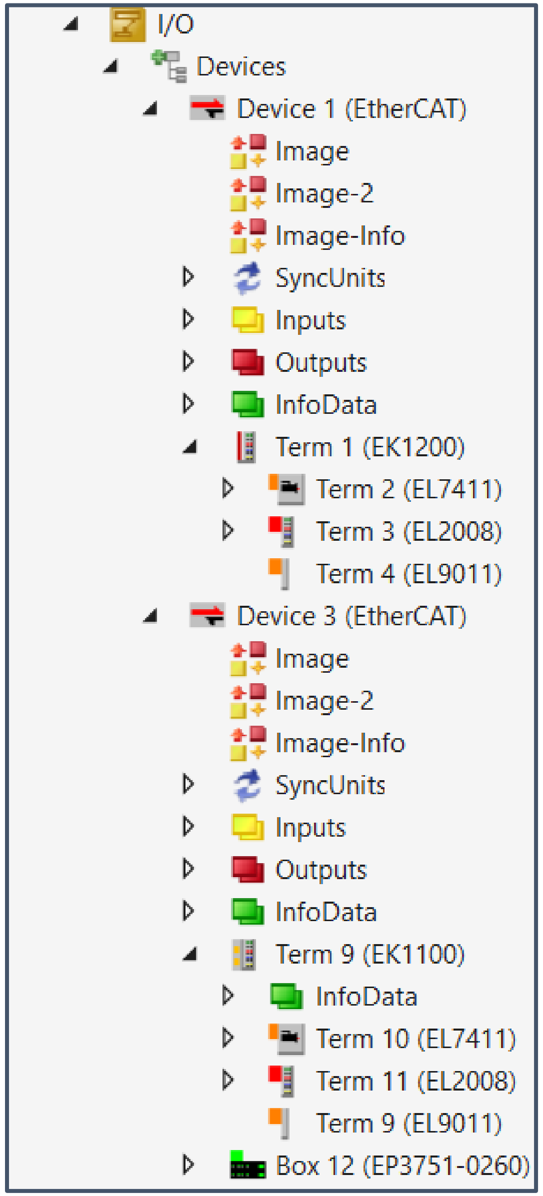
Beckhoff Architecture

PLC Component List







Programming

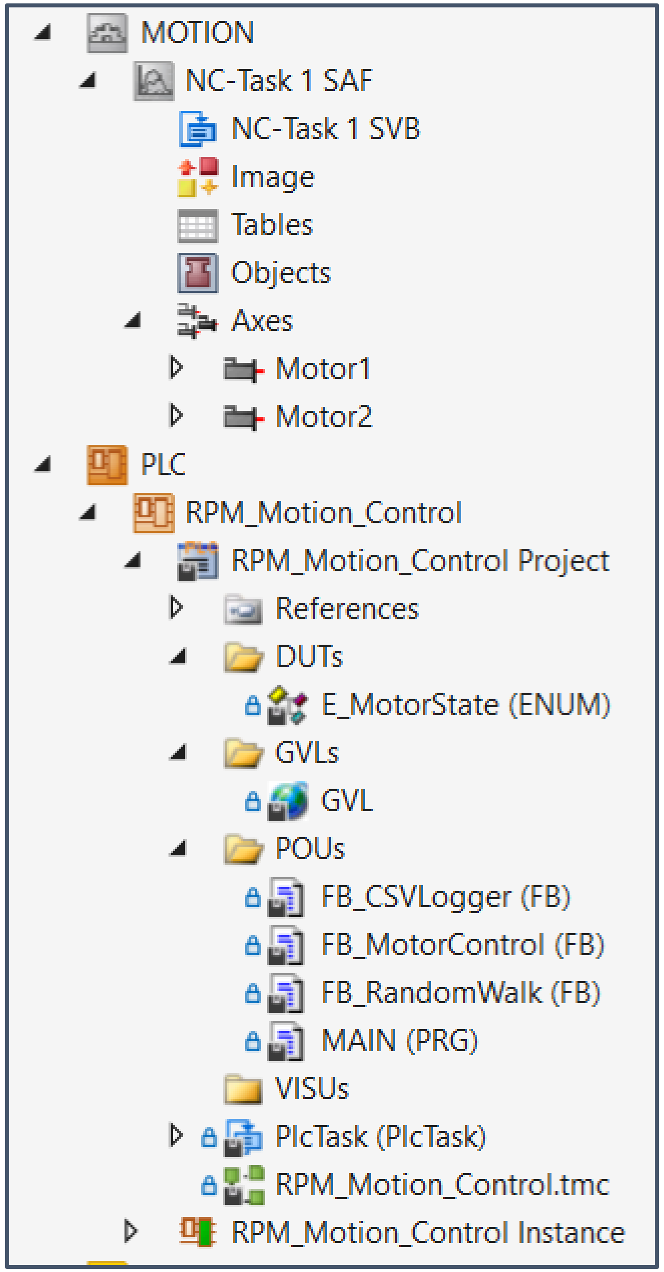
Learning PLC Programming

Program Architecture

- Written on the TwinCAT XAE Shell in Structured Text (ST)
- Three function blocks (FB) + MAIN
    - FB_MotorControl: Controls motor speed, direction
    - FB_CSVLogging: Adds data from motor and IMU to .csv file
    - FB_RandomWalk: Switches direction randomly within a prescribed time frame

Safety Feature Implementation

- Motor power output controlled by output 1 on EL2008, 24V connected to HW-Enable on the EL7411
- If current is too high (stall), the output turns off, motor stops
- Driver output current limited to 5 A
- Motor nominal current a 4.1 A



Electrical and Software Challenges

TwinCAT Realtime delays

IPC

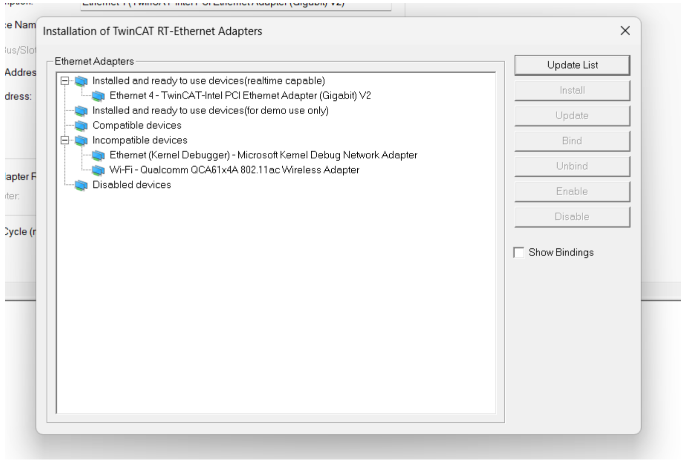
- Did not realize the need for an IPC to run the Beckhoff modules until just before I needed to start programming
- Learned how to operate in the TwinCAT environment through YouTube tutorials while it shipped
- This was my first time writing in ST. To avoid more delays, I relied on AI to write much of the motor control scheme and assist with troubleshooting

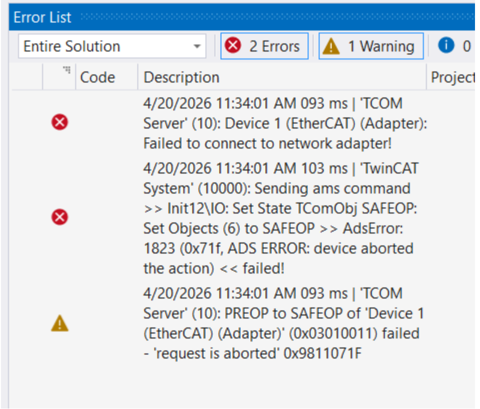
TwinCAT in real time

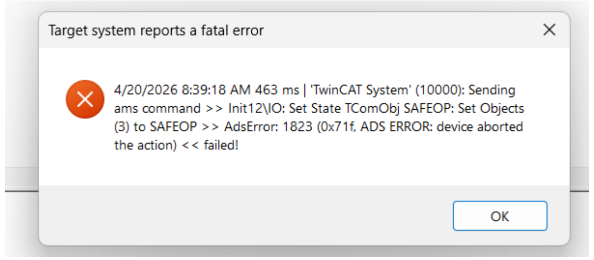
- Achievable by isolating at least a single core on PC for running TwinCAT
- Running in real time is complicated significantly by the driver protocols and security permissions
- Ethernet protocol adapters are very finnicky, and wasted 1-2 weeks of time

CoE Parameters

- > 50 individual parameters that must be set for each motor driver (motor specs, ratings, drive mode configurations, etc.)
- Very time consuming. I made a how-to here





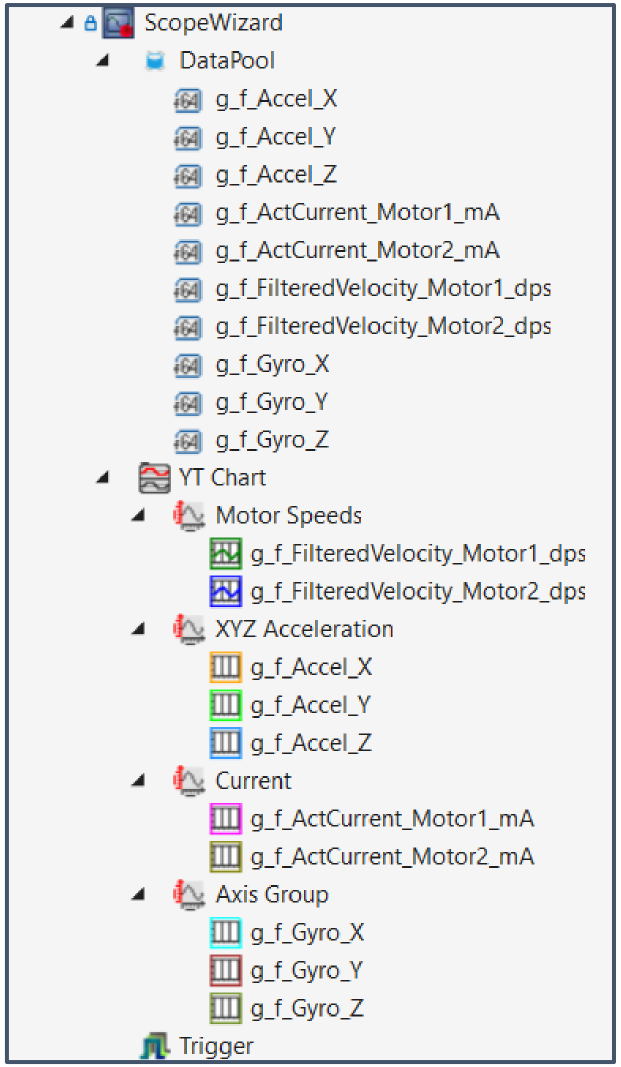


Results

Testing & Results

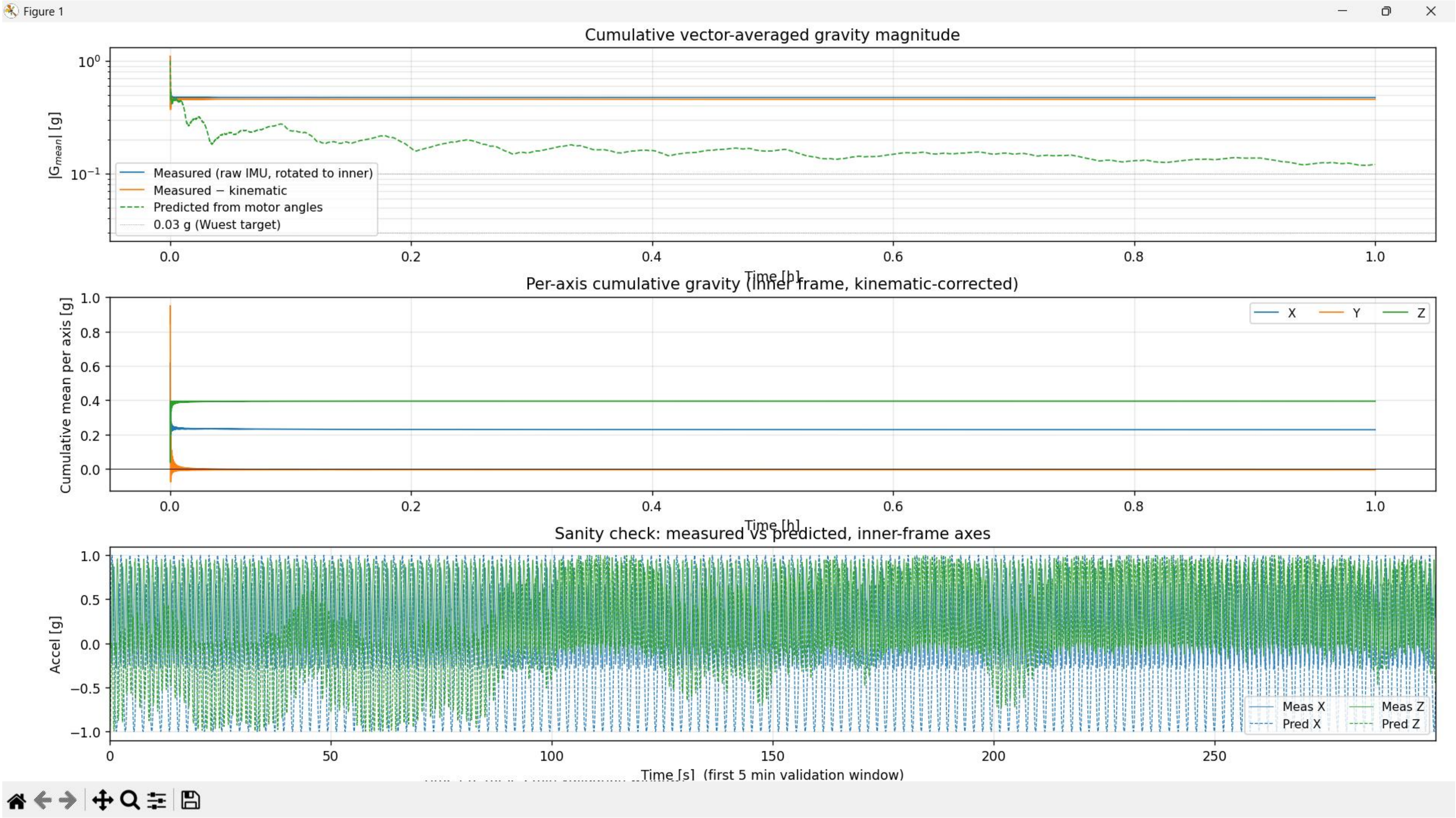
Collecting and Analyzing Data

- Performed various tests in 3D Clinostat (no direction switching) and RPM mode (direction switching randomly every 45-75 sec)
- Data collected
    - X,Y,Z acceleration from accelerometer [g’s]
    - X,Y,Z rotation rates from gyroscope [deg/s]
    - Measured, filtered rotational speeds [deg/s]
    - Motor current [mA]
    - Video of fluorescent tracer particles
- Live data view in Scope View through TwinCAT IDE
- Data exportable to .csv
- Performed x4 1-hr tests at various speed ratios



3D Clinostat Mode – 30:30

Duration 1:00:00



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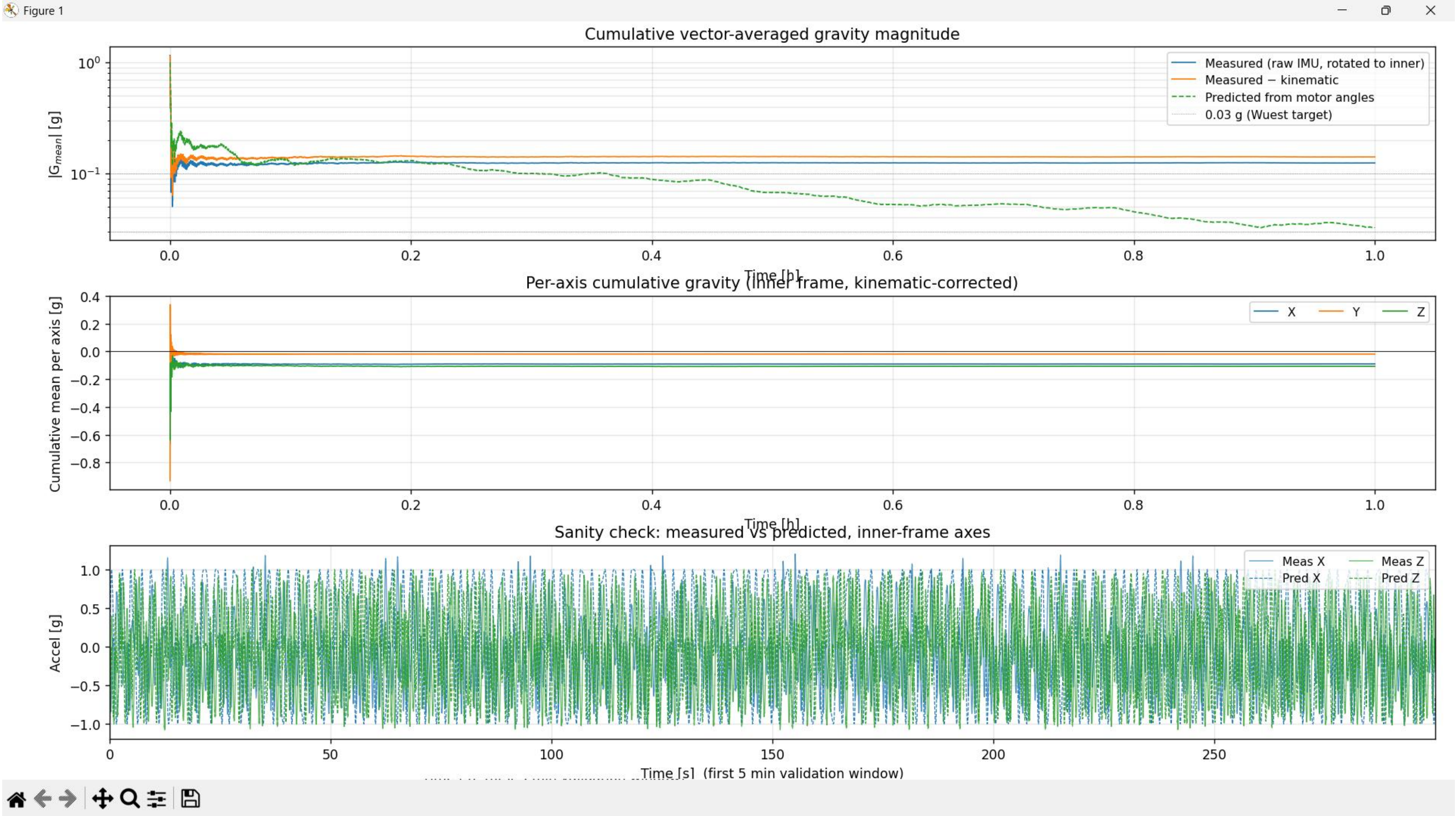
3D Clinostat Mode – 42:25

Duration 1:00:00



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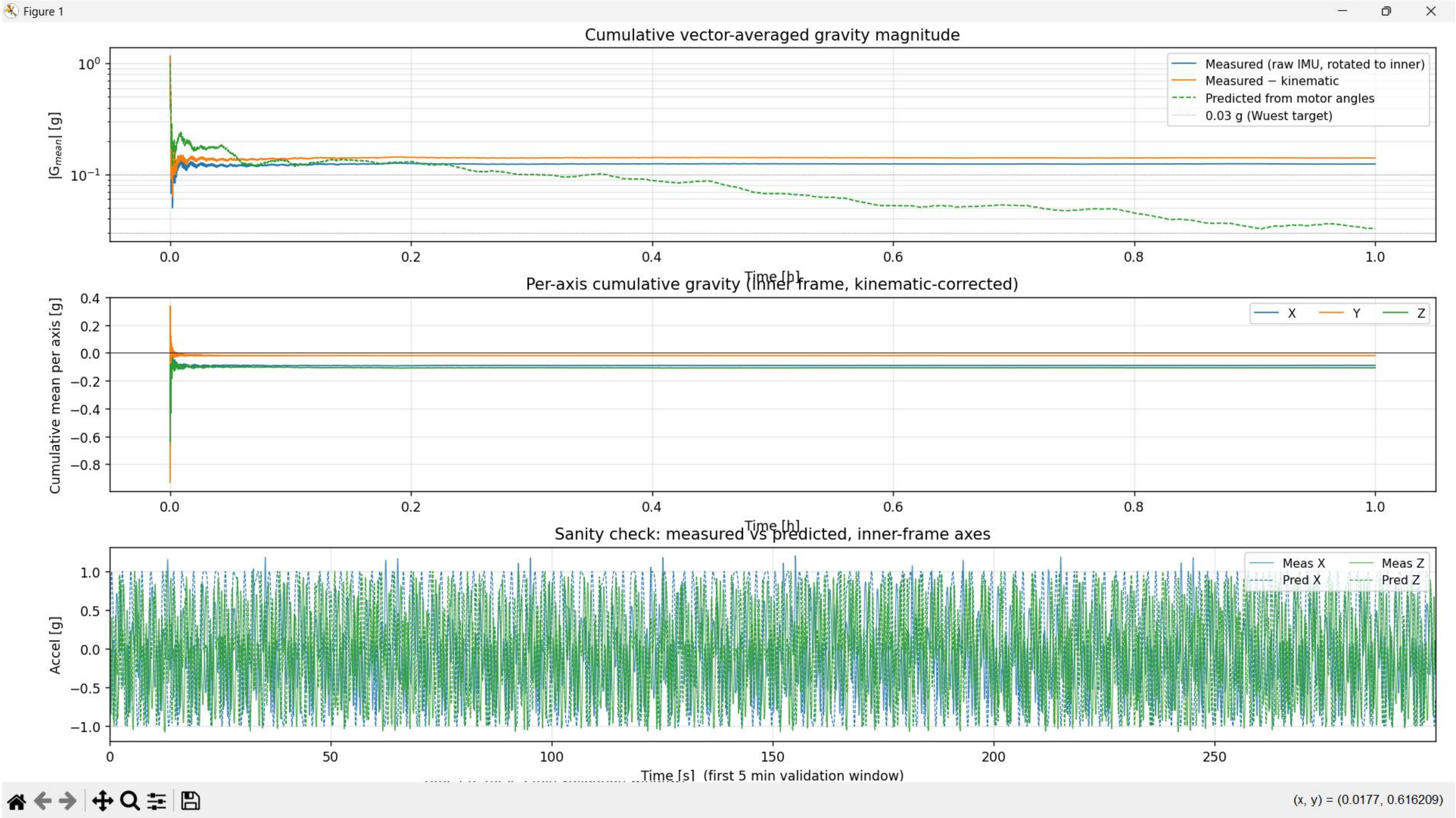
Random Positioning Mode – 42:25

Duration 0:30:00



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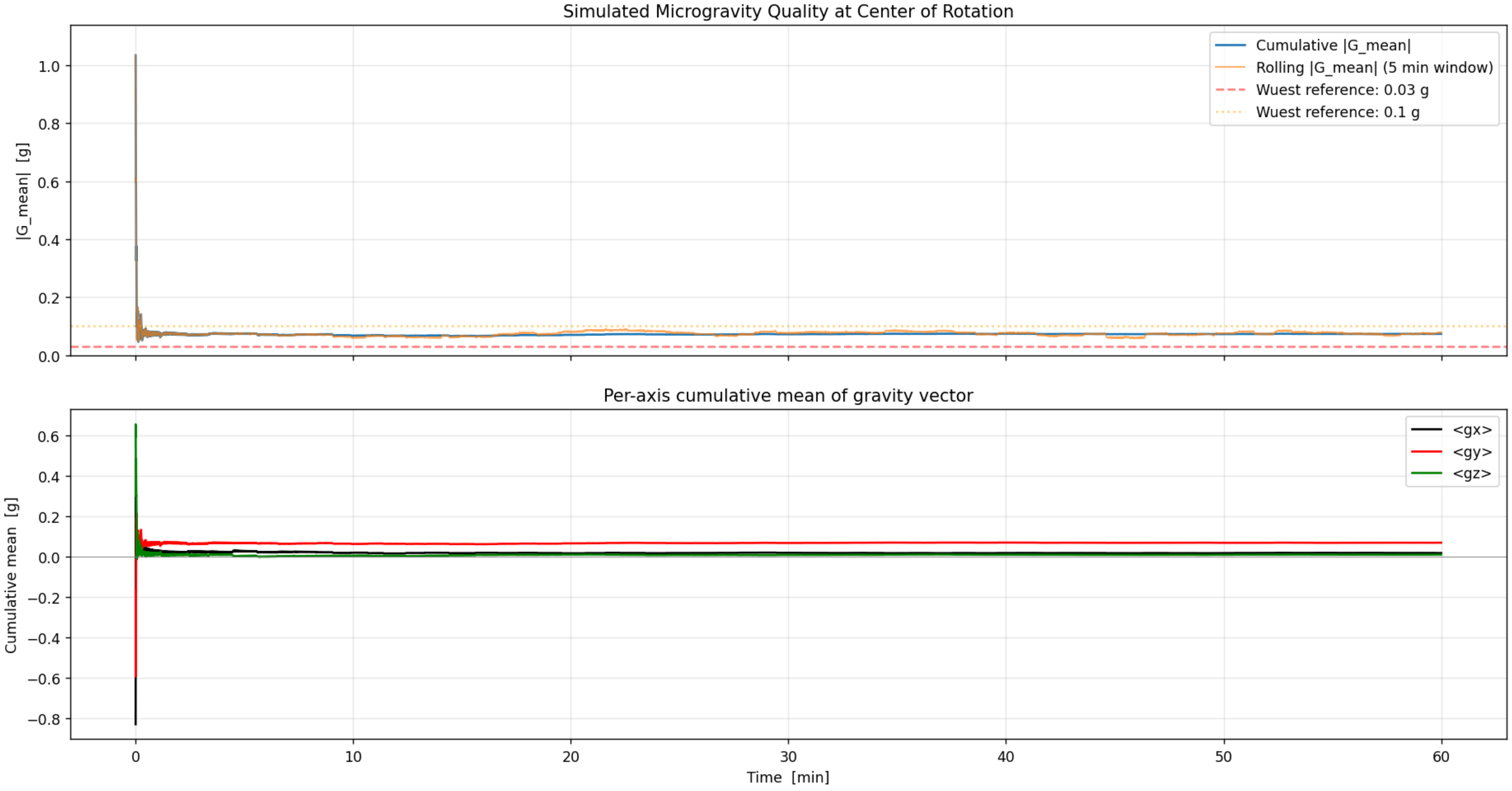
Random Positioning Mode – 42:25

Duration 1:00:00



RPM & 3D Clinostat Testing

Interpreting Results

Test 1 – 3D Clinostat Mode 30:30 speed ratio for 1 hr

- ***|G| = 0.5g***
- Due to the simple ratio between rotational speeds, there was a clear bias in one direction

Test 2 – 3D Clinostat Mode 42:26 speed ratio for 1 hr

- ***|G| = 0.13g***
- The irrational ratio between the rotational speeds helped significantly.

Test 3 – RPM Mode 42:26 speed ratio for 0.5 hr

- ***|G| = 0.11g***
- Switching to RPM mode helped slightly. However, the poorly tuned motors fell into sync again nearly every cycle which negated the effect of the direction switching

Test 4 – RPM Mode 42:26 speed ratio for 1 hr

- ***|G| = 0.07g***
- Tuning motors lowered |G| considerably.
- Potential ways to lower this value are to run longer tests, operate at lower speeds (currently limited by encoder-less feedback),

$$G_{mean} = \sqrt{(G_{X,mean})^2 + (G_{Y,mean})^2 + (G_{Z,mean})^2}$$

$$G_{X,mean} = \frac{\sum_{i=1}^{n} g_{X,I,i}}{n}$$

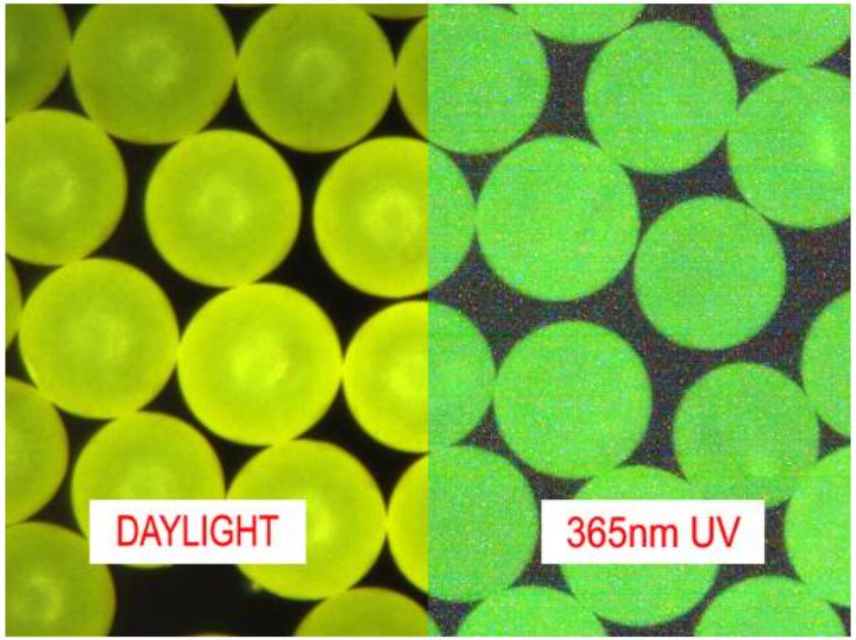
$$G_{Y,mean} = \frac{\sum_{i=1}^{n} g_{Y,I,i}}{n}$$

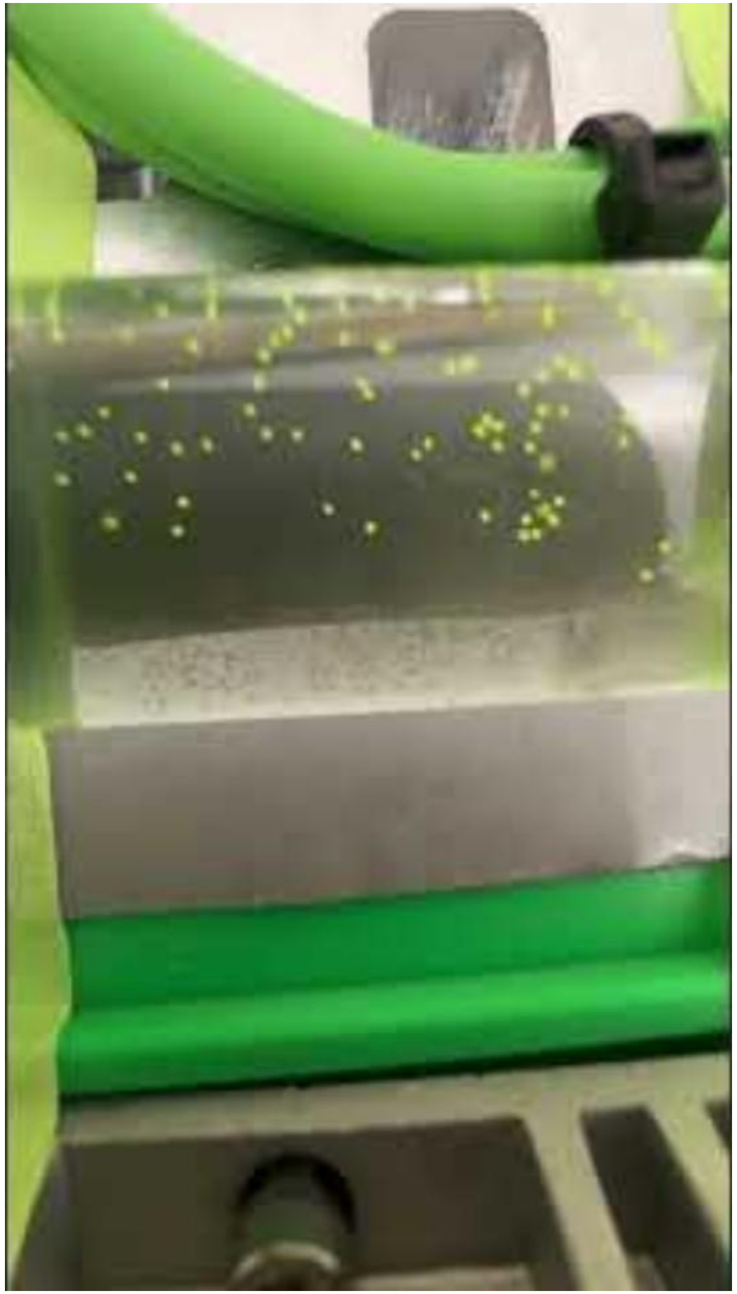
$$G_{Z,mean} = \frac{\sum_{i=1}^{n} g_{Z,I,i}}{n}$$

Tracer Particles Video

Neutrally Buoyant Particles in IPA

- $\Delta\rho$ = 0.22 g/cc (suspended in IPA)
- $D_{part.}$ = 250 µm
- Particles strictly for visualization





Next Steps

Questions & Answers